

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0008

Shroud 100 kV
April 16, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0008. The Interconnection Customer proposes a 93.2 MW natural gas combined cycle facility interconnecting at the Shroud 100 kV substation in the Indiana area. The request was received on January 28, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 5 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,069,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Southeast Chicago Energy Project Substation / Shroud 100 KV line
Contingency: Shroud / Elm Creek 100 KV L
Loading: 133.2% (Rating: 119 MVA)
DFAX: 65.1%
Pre-project loading: 82.0%
2. Facility: Shroud / Elm Creek 100 KV
Contingency: Southeast Chicago Energy Project / Shroud 100 KV
Loading: 112.4% (Rating: 137 MVA)
DFAX: 51.9%
Pre-project loading: 77.1%
3. Facility: Hegewisch / Tower Automotive 100 KV line
Contingency: Tower Automotive / Hegewisch 100 KV line
Loading: 111.6% (Rating: 150 MVA)
DFAX: 25.3%
Pre-project loading: 95.8%
4. Facility: Hegewisch / Tower Automotive 100 KV Line 2
Contingency: Tower Automotive / Hegewisch 100 KV Cir. 1
Loading: 100.8% (Rating: 166 MVA)
DFAX: 25.3%
Pre-project loading: 86.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Southeast Chicago Energy Project Substation / Shroud 100 KV line	Rebuild line with higher-capacity conductor	\$2,810,000	100% (\$2,810,000)
Shroud / Elm Creek 100 KV	Rebuild line with higher-capacity conductor	\$747,000	100% (\$747,000)
Tower Automotive / Hegewisch 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,718,000	100% (\$1,718,000)
Tower Automotive / Hegewisch 100 KV Circuit 2	Reconductor line	\$794,000	100% (\$794,000)
	Total allocated to project		\$6,069,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.